ASSIGNMENT 2
F79SU Survival Models

Semester 2, 2025-26

General instructions

- **Submission deadline.** The deadline for submission is **09:00 am UK time = 1:00 pm Dubai time = 5:00 pm Malaysia time on Friday 27th March 2026** (Week 11); projects may be submitted early. Late projects will be marked in line with the University Coursework Policy, i.e. a standard deduction of 30% if submitted within 5 working days and 0 marks awarded if submitted later. You can, however, submit a mitigating circumstances application, but these may only be applied at the end of the semester. No individual extensions are allowed.

- **Marking.** This assignment will be marked out of 25 and will carry a 25% weight in the final mark for the course.

- **Form of solution.** Your solution to the assignment should take the form of two files: a report and an R script. The report should be prepared using LaTex, Word or other word-processing software, and should contain your answers to the given tasks. The R script should contain the code which is used to produce your answers and plots, and perform any other calculations.

- **Length of report.** Reports should not exceed 8 pages (A4) in length, including graphs and tables, and also including the appendix if you decide to include one and any cited references / bibliography.

- **How to submit.** Assignments must be submitted online on Canvas via Assignments. Online report submissions will be subjected to a plagiarism check through Turnitin. You should also complete the Student Declaration of Authorship quiz at the same time through the appropriate link on Canvas.

- **Use of English.** Answers that are difficult to follow because of poor use of English might lose marks, so it is important to read through your work carefully before submission.

- **Figures and tables.** These are usually necessary to present the results of empirical work clearly. If you include figures and tables in your report, the discussion of their contents should be contained in the main text and not in captions. Please ensure that figures are easy to interpret and use clear labelling and legends.

- **Questions for the Lecturers**: If you have any questions on the assignment, please ask in the discussion board rather than e-mailing lecturers directly. That way all students will see the same responses. The discussion board for this coursework will close at **09:00 am UK time = 1:00 pm Dubai time = 5:00 pm Malaysia time on Thursday 26th March 2026**

- **Collaboration and Plagiarism**: Students are allowed to discuss the methods used with other students, but your submitted project must be **all your own work**. Particularly please note the following:

  - Coursework reports must be written in a student's own words and any code in their coursework must be their own code. If some text or code in the coursework has been taken from other sources, these sources must be properly referenced.
  - Failure to reference work that has been obtained from other sources or to copy the words and/or code of another student is plagiarism and if detected, this will be reported to the School's Discipline Committee. If a student is found guilty of plagiarism, the penalty could involve voiding the course or worse.
  - Students must never give hard or soft copies of their coursework reports or code to another student. Students must always refuse any request from another student for a copy of their report and/or code.
  - Sharing a coursework report and/or code with another student is collusion, and if detected, this will be reported to the School's Discipline Committee. If found guilty of collusion, the penalty could involve voiding the course.

  More information on plagiarism can be found here.

- **Feedback.** We will endeavour to provide feedback by Wednesday 15th April.

Assignment guidelines

General information:

You may make use of all R functions that have been provided in this course, plus standard R functions. However, when you write up your assignment, you should explain what you have done by making reference to the underlying statistics.

You may lose marks if you use other R functions, that have not been used in the class, without further explanation.

Your report must be written independently of other students in your class and all programming work must be your own individual work (except from R code provided in the course).

The following aspects will be considered for marking:

- overall structure of your report;
- R code, calculations and suitability of plots;
- comments, discussion and interpretation of results;
- clarity of writing and general mathematical exposition;
- giving detailed answers, with appropriate explanations and good structure;
- overall conclusions provided.

Guidelines:

- Write up your results in clear English.
- Use clear mathematical formulae to define any distribution, statistical quantity or property that you are analysing. If these are unclear you will lose marks.
- Reread everything you write to look for vagueness, ambiguity, and illogical reasoning. This means taking a step back from the detailed analysis.
- Define clearly all of the assumptions that you are using.

Assignment tasks

Important: each student will use randomly selected data for these tasks. You must follow the following instructions before you start your analysis.

- Run the R script `Assignment2.R` (available on Canvas) with `set.seed(xxxx)`, where `xxxx` is the last four digits of your Heriot-Watt student identification number. For example, if your student ID is H12345678 then your seed would be 5678.
  Failure to do so may result in 0 marks being awarded.
- The R script `Assignment2.R` selects data over the age range 35 to 80, for a particular year, `MyYear`, which is chosen randomly in the range 1950 to 2005 by running the R script.
- Aside from the seed in `set.seed(xxxx)`, do not change any of the code above the line reading 'Please insert your R code after this line.'
  Failure to do so may result in 0 marks being awarded.
- You will need to have the data files `CMI_Deaths.csv` and `CMI_Exposures.csv`, the file `CMI_read.r` and (if you use it) the R-function file `Test_GoF.r` in your working directory. To change your working directory select 'Change dir...' from the 'File' menu if using the R GUI or select "Set Working Directory" then "Choose Directory." from the "Session" menu in RStudio.
  Please do not give references to directories on your own computer (e.g. specifying a directory with setwd) or make use of any R library.

Using the data for the year `MyYear`, you should carry out a Gompertz graduation of the death rates over the age range 35 to 80, using a model with Poisson errors, consider generalisations and compare the fits of your models.

In particular, in your analysis you should do the following:

1. Plot the observed mortality rates on the log scale, i.e. $\log(\hat{\mu}_x)$, against the ages x, where $\hat{\mu}_x$ are the usual crude estimates at age x. Comment on the graph. [2 marks]

2. Fit generalised linear models (GLM) with linear predictors of the form

$$\log(\mu_x) = a_0 + a_1 x \quad \text{(model M1)},$$

$$\log(\mu_x) = b_0 + b_1 x + b_2 (x - \bar{x})^2 \quad \text{(model M2)},$$

$$\log(\mu_x) = c_0 + c_1 x + c_2 (x - \bar{x})^2 + c_3 (x - \bar{x})^3 \quad \text{(model M3)},$$

to the crude rates $\hat{\mu}_x$ assuming Poisson errors for the number of deaths at each age x. Here $\bar{x}$ denotes the average age in the range considered. Ensure you state the values of all coefficients of the expressions for the models fitted. [2 marks]

3. Add the fitted Gompertz line (M1) on the plot of the observed rates $\log(\hat{\mu}_x)$ from Task 1, and comment on your plot. [2 marks]

4. Add the fitted lines for M2 and M3 to the plot of the observed rates $\log(\hat{\mu}_x)$ from Task 3, and comment on your plot. [2 marks]

5. Calculate and plot the Pearson residuals of the three models. Provide initial comments on the overall fit and any other observations on the three models. [3 marks]

6. For each of the three models, perform the following tests on the Pearson residuals, clearly explaining your conclusions: [6 marks]

(a) χ^2 goodness of fit test;

(b) standardised deviations test with 4 equal areas;

(c) sign test;

(d) change of signs test;

(e) runs test;

(f) lag-1 serial correlations test.

7. The Akaike Information Criterion (AIC) of a fitted GLM is expressed as

$$\text{AIC} = 2k - 2\ln(\hat{L})$$

where $\hat{L}$ is the maximised value of the likelihood function for the model and k is the number of estimated parameters in the model. The AIC can be used to compare fitted models, with models having smaller AIC being preferred in general. Obtain the value of the AIC for each of the three models fitted in Task 2, and comment. You can obtain the AIC for a fitted GLM by using "$aic" [2 marks]

8. Based on all of the above, comment on the overall suitability of the three models, explaining which model would be preferred if one model had to be used for applications in insurance and pensions. [3 marks]

A further **3 marks** are available for the overall structure, flow and presentation of your report.

[Assignment total: 25 marks]

END OF THE ASSIGNMENT